Android Forensic Examination Report

acceptance

Generated 2026-09-22T07:49:42.962Z - android-forensics 0.1.0

Review 1 + Review 2 - deleted-data recovery performed. Evidence integrity, filesystem analysis, SQLite discovery, extraction of ACTIVE artifacts, recovery of deleted and superseded SQLite records (freelist, unallocated space, WAL and rollback-journal analysis), raw pattern carving, cross-artifact correlation, deterministic confidence scoring, validation, timeline and reporting.

1. Case information

Case name	acceptance
Case ID	cc7be14b9e3b445b812860cdb0c1b5be
Description	Phase 17 end-to-end acceptance: synthetic_evidence/ with genuine deletions
Case created (UTC)	2026-09-22T07:48:19.958Z
Report generated (UTC)	2026-09-22T07:49:42.962Z

Evidence at a glance:

- ACTIVE DATA: 346 (records present in live databases)
- RECOVERED DELETED DATA: 8,218 (reconstructed with a confidence explanation)
- PARTIAL RECOVERIES: 26 (reconstructed; missing fields are UNKNOWN, never guessed)
- CARVED CANDIDATES: 177 (unconfirmed pattern hits, never ACTIVE)
- UNRECOVERABLE DATA: 4 (proven deleted, content lost)

2. Examiner and tool information

Examiner	B.Tech Capstone Acceptance Run
Tool	android-forensics 0.1.0
Python	3.12.13
SQLite library	3.53.3
Platform	macOS-26.5.2-arm64-arm-64bit

3. Evidence sources

Source	Path	Type	Files	Complete
s1	/Users/harshamithran/Documents/cf/synthetic_evidence/a	directory	16	yes

4. Evidence hashes

SHA-256 is the integrity anchor. The complete list is in evidence_manifest.json, report.json and csv/evidence_files.csv.

Path	Kind	Size	SHA-256
data/data/com.android.chrome/app_chrome/Default/Hi	sqlite_db	32,768	47dcf5590d066ede9f15dce2fbebbd33b9edc49fd9134cdbf
data/data/com.android.providers.contacts/databases/c	sqlite_db	28,672	cb2f416b05fc4d68ff33420ba5d12865f6bd16f30db99361e
data/data/com.android.providers.contacts/databases/c	sqlite_shm	32,768	dff5a248abe5b150affd075255cd60273da04d41df132f95f1
data/data/com.android.providers.contacts/databases/c	sqlite_wal	20,632	bbb08ac2ef74c53327dc16870dfb59401b0ffed03457f8105
data/data/com.android.providers.contacts/databases/c	sqlite_db	36,864	3acaf1ab80e5e1452b2c522285a5aad656e8b1bce923ee5
data/data/com.android.providers.telephony/databases	sqlite_db	36,864	37cb16c64bf3c92c93a2c3401729598c7e224a9f0521d730
data/data/com.example.corrupt/databases/broken.db	sqlite_db	29,437	3a7c448a6f7df10337574c2a8ec5fa6c9c24c595dea61ee8
data/data/com.example.notes/databases/notes.db	sqlite_db	16,384	37116012f68b0ee30748098a1df4ee8c5075b2ff3eca9dd5
data/data/com.example.notes/databases/pending.db	sqlite_db	573,440	1f43882e4fdea9db27ed0c1cc646bf16640aa92a47b872c8
data/data/com.example.notes/databases/pending.db-j	sqlite_journal	570,968	73e725eeef7353704e7fe229c7fc27f12345ac6582806a34
data/data/com.example.notes/shared_prefs/com.exa	xml	266	6b35fd757303fe7c8b4c387968ccda04de46b3df46d05410
data/data/com.example.secure/databases/vault.db	encrypted_suspec	16,384	946b1e82db5dae2cc206cc2e74c98337ac3bf378bee7322
data/system/packages.xml	xml_binary	260	e2a17c648f33ff85f736f402d79ac47eae6c7614a38972820
data/system/users/0.xml	xml	90	4e57224c833274b931d1358de367b27cbf9406356c74947
sdcard/DCIM/Camera/IMG_20240301_101500.jpg	image	2,061	1431b5a98c2db460749cc62d5cf25b31ba8afc0061e8b368
storage/emulated/0/Pictures/screenshot.png	image	1,032	3ba66c32c8342c21df666d1d1e00a1a94e20e0c135cd529f

5. Acquisition summary

Files hashed 16

Write-protected working copies 13

Acquisition complete yes

Detected layout android_root

SQLite databases:

Database	Status	Open	Integrity	Tables	Artifacts
data/data/com.android.chrome/app_chrome/D	VALID	ok	VALID	6	162
data/data/com.android.providers.contacts/dat	VALID	ok	VALID	3	60
data/data/com.android.providers.contacts/dat	VALID	ok	VALID	7	34
data/data/com.android.providers.telephony/da	VALID	ok	VALID	4	121
data/data/com.example.corrupt/databases/bro	TRUNCATED	unopenable	UNVERIFIED	0	207
data/data/com.example.notes/databases/note	VALID	ok	VALID	2	0
data/data/com.example.notes/databases/pend	VALID	ok	VALID	1	8000
data/data/com.example.secure/databases/vau	ENCRYPTED_SU	encrypted_su	UNVERIFIED	0	0

6. Artifact summary

Type	Count
app_summary	6
browser_download	5
browser_visit	156
call	60
contact	30
generic_record	8,213
sms	120

7. Recovered deleted data

Records reconstructed from freed cells, released pages, stale page images and WAL/journal history. A column the bytes did not determine is never guessed.

Status	Table	Rowid	Method	Confidence	Explanation
RECOVERED	meta	1	RAW_PAGE_SC	LOW	LOW: full page-level reconstruction (the whole cell survived); limite
RECOVERED	mimetypes	4	RAW_PAGE_SC	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s
RECOVERED	mimetypes	3	RAW_PAGE_SC	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s
RECOVERED	mimetypes	2	RAW_PAGE_SC	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s
RECOVERED	mimetypes	1	RAW_PAGE_SC	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s
RECOVERED	android_metad	1	RAW_PAGE_SC	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s
RECOVERED	sms	28	UNALLOCATED	HIGH	HIGH: full page-level reconstruction (the whole cell survived), sche
RECOVERED	sms	15	UNALLOCATED	HIGH	HIGH: full page-level reconstruction (the whole cell survived), sche
RECOVERED	sms	6	UNALLOCATED	HIGH	HIGH: full page-level reconstruction (the whole cell survived), sche
RECOVERED	sms	4	UNALLOCATED	HIGH	HIGH: full page-level reconstruction (the whole cell survived), sche
RECOVERED	sms	68	UNALLOCATED	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s
RECOVERED	log	42	RAW_PAGE_SC	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s
RECOVERED	log	41	RAW_PAGE_SC	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s
RECOVERED	log	40	RAW_PAGE_SC	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s
RECOVERED	log	39	RAW_PAGE_SC	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s
RECOVERED	log	38	RAW_PAGE_SC	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s
RECOVERED	log	37	RAW_PAGE_SC	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s
RECOVERED	log	36	RAW_PAGE_SC	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s
RECOVERED	log	35	RAW_PAGE_SC	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s
RECOVERED	log	34	RAW_PAGE_SC	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s
RECOVERED	log	33	RAW_PAGE_SC	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s
RECOVERED	log	32	RAW_PAGE_SC	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s
RECOVERED	log	31	RAW_PAGE_SC	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s
RECOVERED	log	30	RAW_PAGE_SC	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s
RECOVERED	log	29	RAW_PAGE_SC	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s
RECOVERED	log	28	RAW_PAGE_SC	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s
RECOVERED	log	27	RAW_PAGE_SC	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s
RECOVERED	log	26	RAW_PAGE_SC	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s
RECOVERED	log	25	RAW_PAGE_SC	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s
RECOVERED	log	24	RAW_PAGE_SC	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s
RECOVERED	log	23	RAW_PAGE_SC	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s
RECOVERED	log	22	RAW_PAGE_SC	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s
RECOVERED	log	21	RAW_PAGE_SC	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s
RECOVERED	log	20	RAW_PAGE_SC	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s
RECOVERED	log	19	RAW_PAGE_SC	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s
RECOVERED	log	18	RAW_PAGE_SC	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s
RECOVERED	log	17	RAW_PAGE_SC	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s
RECOVERED	log	16	RAW_PAGE_SC	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s
RECOVERED	log	15	RAW_PAGE_SC	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s
RECOVERED	log	14	RAW_PAGE_SC	MEDIUM	MEDIUM: full page-level reconstruction (the whole cell survived), s

Showing the first 40 of 8218 rows. The complete data is in report.json and the CSV files.

Recovery-potential indicators:

Database	Freelist pages	WAL frame slots	Notes
data/data/com.android.chrome/app_chrome/D	0		no indicator present in header or sidecars (this does not prove
data/data/com.android.providers.contacts/dat	0	5	WAL present (5 frame slot(s)) ? may hold superseded page im
data/data/com.android.providers.contacts/dat	0		no indicator present in header or sidecars (this does not prove
data/data/com.android.providers.telephony/da	0		no indicator present in header or sidecars (this does not prove
data/data/com.example.corrupt/databases/bro	0		database not readable by SQLite ? a candidate for Review 2 s
data/data/com.example.notes/databases/note	0		no indicator present in header or sidecars (this does not prove
data/data/com.example.notes/databases/pend	0		valid-magic rollback journal ? holds pre-transaction page imag

Unrecoverable data:

Source	Region	Reason
/Users/harshamithran/Documents/cf/cases/ac	freeblock	zero-filled: the content was erased (secure_delete, or overwritten with zeros)
/Users/harshamithran/Documents/cf/cases/ac	freeblock	the bytes are not a record of any known table (partly overwritten, or another st
/Users/harshamithran/Documents/cf/cases/ac	freeblock	zero-filled: the content was erased (secure_delete, or overwritten with zeros)
/Users/harshamithran/Documents/cf/cases/ac	freeblock	the bytes are not a record of any known table (partly overwritten, or another st

8. Active data

Records present in the live database view. Full provenance for each row is in report.json / the CSV export.

SMS messages (108)

ID	Time (UTC)	Direction	Address	Body	Read	Thread
236	2024-03-02T13:02:	RECEIVED	+12025550120	Dinner at my place	True	5
292	2024-03-03T05:46:	RECEIVED	56161	Package out for deli	True	12
281	2024-03-03T15:44:	RECEIVED	+12025550108	Can you call me wh	False	7
246	2024-03-03T18:23:	DRAFT		Meeting moved to r	True	13
321	2024-03-05T09:29:	SENT	+12025550114	Reminder: appoint	True	34
272	2024-03-06T03:04:	SENT	+12025550120	Bank alert: Rs 6494	True	5
249	2024-03-07T01:56:	SENT	+12025550104	Package out for deli	True	10
337	2024-03-08T11:06:	RECEIVED	+12025550106	Payment of Rs 573	True	4
315	2024-03-08T23:37:	RECEIVED	+12025550104	Package out for deli	False	10
340	2024-03-10T10:08:	RECEIVED	+12025550100	Payment of Rs 638	True	19
280	2024-03-10T17:25:	SENT	+12025550124	Package out for deli	True	2
267	2024-03-11T06:28:	RECEIVED	AX-BANKOS	Please send the re	True	22
253	2024-03-11T11:01:	SENT	+12025550107	Can you call me wh	True	15
247	2024-03-11T16:15:	RECEIVED	+12025550103	Please send the re	True	14
277	2024-03-11T18:35:	RECEIVED	+12025550129	Payment of Rs 731	True	11
310	2024-03-13T15:58:	SENT	+12025550114	Thanks a lot!	True	34
240	2024-03-16T20:11:	RECEIVED	+12025550108	Dinner at my place	False	7
313	2024-03-17T22:19:	RECEIVED	72000	Payment of Rs 544	True	23
245	2024-03-18T08:24:	RECEIVED	56161	Meeting moved to r	True	12
304	2024-03-18T10:41:	SENT	+12025550120	Package out for deli	True	5
271	2024-03-18T12:00:	RECEIVED	JD-DELIVR	Can you call me wh	True	26
324	2024-03-20T01:54:	RECEIVED	+12025550107	Thanks a lot!	True	15
308	2024-03-22T02:01:	RECEIVED	+12025550124	Dinner at my place	True	2
297	2024-03-22T10:09:	RECEIVED	+12025550123	Meeting moved to r	True	20
288	2024-03-23T21:50:	RECEIVED	+12025550116	Bring the document	True	31
284	2024-03-24T19:31:	DRAFT		Reminder: appoint	True	13
300	2024-03-26T09:51:	RECEIVED	+12025550115	Call me back asap	True	29
257	2024-03-27T10:10:	SENT	+12025550112	Good night :)	True	21
331	2024-03-27T10:14:	SENT	+12025550122	Are we still meeting	True	6
317	2024-03-29T15:11:	RECEIVED	+12025550121	Payment of Rs 503	False	27
298	2024-03-30T12:08:	RECEIVED	+12025550128	Bring the document	False	33
244	2024-03-30T14:04:	RECEIVED	+12025550104	Ok	True	10
282	2024-04-01T00:19:	SENT	+12025550109	Reminder: appoint	True	3
335	2024-04-01T06:24:	RECEIVED	JD-DELIVR	Thanks a lot!	False	26
326	2024-04-01T10:57:	RECEIVED	+12025550117	Your OTP is 88185	False	1
261	2024-04-01T11:02:	SENT	+12025550109		True	3
301	2024-04-02T16:17:	RECEIVED	+12025550126	Happy birthday! Ha	True	35
276	2024-04-03T16:16:	SENT	+12025550113	Running late, reach	True	28
239	2024-04-04T01:16:	RECEIVED	+12025550119	Thanks a lot!	True	8
311	2024-04-04T13:37:	SENT	+12025550118	Lecture notes uploa	True	24

Showing the first 40 of 108 rows. The complete data is in report.json and the CSV files.

Call log (60)

ID	Time (UTC)	Type	Number	Duration (s)	Cached name	Presentation
198	2024-03-01T23:30:	MISSED	+12025550129	0	Sneha Gupta	ALLOWED
147	2024-03-02T14:16:	INCOMING	+12025550108	1651	Dev Kulkarni	ALLOWED
164	2024-03-05T20:06:	OUTGOING	+12025550102	757	Rohan Verma	ALLOWED
170	2024-03-06T12:09:	OUTGOING	+12025550125	831		ALLOWED
183	2024-03-06T19:42:	REJECTED	+12025550113	0		ALLOWED
182	2024-03-06T20:46:	REJECTED	+12025550114	0	Neel Iyer	ALLOWED
160	2024-03-08T06:58:	MISSED	+12025550112	0	Karan Menon	ALLOWED
163	2024-03-09T06:33:	INCOMING	+12025550104	204	Kabir Rao	ALLOWED
172	2024-03-10T20:25:	MISSED	+12025550125	0	Mira Singh	ALLOWED
165	2024-03-11T16:24:	MISSED	+12025550103	0		ALLOWED
159	2024-03-12T15:08:	INCOMING	+12025550103	1607	Isha Nair	ALLOWED
144	2024-03-13T02:17:	MISSED	+12025550100	0	Aarav Patel	ALLOWED
202	2024-03-15T09:56:	INCOMING	+12025550102	230		ALLOWED
171	2024-03-15T23:32:	OUTGOING	+12025550104	461	Kabir Rao	ALLOWED
149	2024-03-16T00:17:	REJECTED	(private/unknown)	0		RESTRICTED
200	2024-03-16T01:23:	INCOMING	+12025550117	1401	Diya Khan	ALLOWED
190	2024-03-16T08:12:	INCOMING	+12025550101	824	Meera Shah	ALLOWED
148	2024-03-18T21:56:	OUTGOING	+12025550124	1143		ALLOWED
143	2024-03-19T20:43:	OUTGOING	+12025550110	1686	Arjun Kapoor	ALLOWED
186	2024-03-21T06:44:	REJECTED	+12025550121	0		ALLOWED
201	2024-03-24T05:04:	OUTGOING	+12025550126	1688	Imran Reddy	ALLOWED
167	2024-03-25T18:10:	INCOMING	+12025550104	551	Kabir Rao	ALLOWED
180	2024-03-25T20:27:	INCOMING	+12025550104	1233		ALLOWED
174	2024-03-26T09:45:	OUTGOING	+12025550124	1098	Yash Rao	ALLOWED
194	2024-03-26T23:17:	MISSED	+12025550103	0		ALLOWED
158	2024-04-03T01:32:	INCOMING	+12025550124	390	Yash Rao	ALLOWED
189	2024-04-03T16:19:	OUTGOING	+12025550100	417		ALLOWED
151	2024-04-04T15:40:	REJECTED	+12025550106	0	Vikram Reddy	ALLOWED
161	2024-04-05T19:25:	OUTGOING	+12025550122	708		ALLOWED
150	2024-04-08T02:29:	OUTGOING	+12025550126	168	Imran Reddy	ALLOWED
157	2024-04-10T02:07:	OUTGOING	+12025550126	567	Imran Reddy	ALLOWED
153	2024-04-10T19:59:	REJECTED	+12025550127	0	Anika Jain	ALLOWED
175	2024-04-12T09:41:	REJECTED	+12025550123	0	Kiara Nair	ALLOWED
154	2024-04-12T12:34:	INCOMING	+12025550127	1462	Anika Jain	ALLOWED
195	2024-04-19T10:25:	OUTGOING	+12025550113	293	Priya Mehta	ALLOWED
187	2024-04-20T04:25:	INCOMING	+12025550125	24	Mira Singh	ALLOWED
179	2024-04-20T21:35:	MISSED	+12025550111	0	Zoya Sharma	ALLOWED
193	2024-04-22T10:04:	OUTGOING	+12025550117	647	Diya Khan	ALLOWED
197	2024-04-29T20:43:	MISSED	+12025550117	0	Diya Khan	ALLOWED
145	2024-05-01T13:56:	REJECTED	+12025550108	0	Dev Kulkarni	ALLOWED

Showing the first 40 of 60 rows. The complete data is in report.json and the CSV files.

Contacts (30)

ID	Last updated (UTC	Name	Phones	Emails	Account	Application flag

ID	Last updated (UTC	Name	Phones	Emails	Account	Application flag
213	2024-03-01T17:04:	Arjun Kapoor	(202) 555-0110	arjun.kapoor@exa	owner@example.co	
221	2024-03-02T07:06:	Omar Bose	(202) 555-0118		owner@example.co	
227	2024-03-02T09:44:	Yash Rao	(202) 555-0124		owner@example.co	
219	2024-03-08T02:44:	Ravi Pillai	(202) 555-0116	ravi.pillai@example	owner@example.co	
215	2024-03-08T04:15:	Karan Menon	(202) 555-0112	karan.menon@exa	owner@example.co	
206	2024-03-08T13:54:	Isha Nair	(202) 555-0103	isha.nair@example.	owner@example.co	
207	2024-03-09T12:16:	Kabir Rao	(202) 555-0104		owner@example.co	
209	2024-03-16T14:32:	Vikram Reddy	(202) 555-0106		owner@example.co	
232	2024-03-20T02:28:	Sneha Gupta	(202) 555-0129	sneha.gupta@exa	owner@example.co	FLAGGED DELET
225	2024-03-20T07:48:	Farhan Verma	(202) 555-0122	farhan.verma@exa	owner@example.co	
223	2024-03-24T14:32:	Sameer Patel	(202) 555-0120	sameer.patel@exa	owner@example.co	
217	2024-03-24T22:07:	Neel Iyer	(202) 555-0114	neel.iyer@example.	owner@example.co	
205	2024-03-25T15:56:	Rohan Verma	(202) 555-0102	rohan.verma@exa	owner@example.co	
231	2024-03-25T23:35:	Rahul Kulkarni	(202) 555-0128		owner@example.co	FLAGGED DELET
203	2024-03-27T01:18:	Aarav Patel	(202) 555-0100		owner@example.co	
220	2024-03-29T23:24:	Diya Khan	(202) 555-0117		owner@example.co	
230	2024-03-31T15:46:	Anika Jain	(202) 555-0127	anika.jain@exampl	owner@example.co	FLAGGED DELET
214	2024-04-05T09:00:	Zoya Sharma	(202) 555-0111	zoya.sharma@exa	owner@example.co	
229	2024-04-06T20:32:	Imran Reddy	(202) 555-0126	imran.reddy@exam	owner@example.co	
208	2024-04-07T09:24:	Nisha Singh	(202) 555-0105		owner@example.co	
216	2024-04-08T12:30:	Priya Mehta	(202) 555-0113	priya.mehta@exam	owner@example.co	
204	2024-04-11T03:01:	Meera Shah	(202) 555-0101	meera.shah@exam	owner@example.co	
210	2024-04-11T07:35:	Anaya Jain	(202) 555-0107	anaya.jain@exampl	owner@example.co	
224	2024-04-11T17:45:	Riya Shah	(202) 555-0121	riya.shah@example	owner@example.co	
211	2024-04-12T01:47:	Dev Kulkarni	(202) 555-0108	dev.kulkarni@exam	owner@example.co	
212	2024-04-20T15:41:	Tara Gupta	(202) 555-0109	tara.gupta@exampl	owner@example.co	
222	2024-04-23T11:15:	Leela Joshi	(202) 555-0119		owner@example.co	
218	2024-04-24T07:20:	Sara Das	(202) 555-0115	sara.das@example	owner@example.co	
228	2024-04-27T06:13:	Mira Singh	(202) 555-0125	mira.singh@exampl	owner@example.co	
226	2024-04-28T13:24:	Kiara Nair	(202) 555-0123	kiara.nair@exampl	owner@example.co	

Browser visits (138)

ID	Time (UTC)	Title	URL	Transition	Visits
30	2024-03-01T21:11:10.	Video #38	https://video.example.c	TYPED	5
136	2024-03-01T22:40:54.	Forum Thread #29	https://forum.example.	url-level record	1
66	2024-03-02T11:26:36.	Weather #27	https://weather.exampl	RELOAD	6
35	2024-03-02T17:06:11.	Weather #27	https://weather.exampl	TYPED	6
126	2024-03-03T18:24:48.	Student Portal #36	https://university.exam	RELOAD	7
67	2024-03-03T19:54:30.	Inbox #33	https://mail.example.co	TYPED	6
104	2024-03-04T18:34:47.	World News	https://news.example.o	TYPED	5
20	2024-03-04T21:47:24.	Shared Doc #20	https://docs.example.c	RELOAD	6
83	2024-03-06T01:18:20.			RELOAD	
11	2024-03-06T11:55:49.	Video	https://video.example.c	LINK	3
75	2024-03-07T06:04:39.	Wikipedia #11	https://www.wikipedia.	RELOAD	4
1	2024-03-07T15:29:42.			RELOAD	
45	2024-03-07T16:44:51.	World News #32	https://news.example.o	RELOAD	4

ID	Time (UTC)	Title	URL	Transition	Visits
7	2024-03-07T21:44:20.	World News	https://news.example.o	LINK	5
81	2024-03-08T15:47:54.	Shared Doc #20	https://docs.example.c	RELOAD	6
16	2024-03-10T04:23:54.			GENERATED	
125	2024-03-10T07:05:27.	Shopping Cart #35	https://shop.example.c	LINK	5
106	2024-03-11T03:09:58.	Maps #14	https://maps.example.	TYPED	2
17	2024-03-11T18:05:49.	Forum Thread	https://forum.example.	RELOAD	5
27	2024-03-11T22:52:23.	Inbox #13	https://mail.example.co	GENERATED	2
9	2024-03-13T00:35:38.	World News	https://news.example.o	TYPED	5
32	2024-03-13T11:32:18.	Student Portal #26	https://university.exam	RELOAD	5
78	2024-03-15T15:44:58.	Maps	https://maps.example.	TYPED	5
112	2024-03-17T05:59:24.	Inbox #33	https://mail.example.co	GENERATED	6
115	2024-03-17T15:36:37.	Video #28	https://video.example.c	TYPED	5
107	2024-03-18T01:53:06.	World News #12	https://news.example.o	TYPED	5
36	2024-03-18T11:01:07.	Student Portal #36	https://university.exam	GENERATED	7
23	2024-03-19T03:37:29.	Maps	https://maps.example.	LINK	5
57	2024-03-19T04:55:48.	Shared Doc #30	https://docs.example.c	GENERATED	3
117	2024-03-20T00:25:17.			LINK	
5	2024-03-20T05:35:29.	Shopping Cart	https://shop.example.c	TYPED	7
68	2024-03-20T16:41:04.	Shopping Cart #25	https://shop.example.c	GENERATED	2
59	2024-03-20T21:56:20.	Wikipedia #11	https://www.wikipedia.	LINK	4
63	2024-03-21T15:14:14.	Forum Thread #19	https://forum.example.	GENERATED	5
97	2024-03-22T08:23:01.	Weather #27	https://weather.exampl	TYPED	6
113	2024-03-22T19:58:51.	Wikipedia	https://www.wikipedia.	TYPED	3
82	2024-03-23T04:22:34.	Video	https://video.example.c	LINK	3
34	2024-03-23T11:46:56.	Inbox #33	https://mail.example.co	TYPED	6
73	2024-03-26T00:28:10.	Maps #24	https://maps.example.	RELOAD	4
133	2024-03-27T06:02:55.	Shopping Cart #15	https://shop.example.c	LINK	3

Showing the first 40 of 138 rows. The complete data is in report.json and the CSV files.

Browser downloads (4)

ID	Time (UTC)	File	Source page	Bytes	State
140	2024-03-10T00:28:20.	file2.pdf	https://docs.example.c	20000	COMPLETE
142	2024-03-31T19:13:42.	file5.pdf	https://docs.example.c	50000	COMPLETE
139	2024-04-02T00:58:13.	file1.pdf	https://docs.example.c	10000	COMPLETE
141	2024-05-08T09:01:47.	file4.pdf	https://docs.example.c	40000	COMPLETE

Application summary (6)

ID	Package	Application	Files	Databases	Last file change (U	Preference keys
341	com.android.chrom	Google Chrome	1	1	2024-06-04T08:00:	0
342	com.android.provid	Android Contacts P	4	2	2024-06-04T08:02:	0
343	com.android.provid	Android Telephony	1	1	2024-06-04T08:03:	0
344	com.example.corru		1	1	2024-06-04T08:03:	0
345	com.example.notes		4	2	2024-06-04T08:06:	4
346	com.example.secur		1	0	2024-06-04T08:06:	0

9. Partial recoveries

Table	Method	Unknown columns	Confidence	Explanation
downloads	FREEBLOCK	id	MEDIUM	MEDIUM: reconstruction from a freed cell whose
urls	FREEBLOCK	id	MEDIUM	MEDIUM: reconstruction from a freed cell whose
urls	FREEBLOCK	id	MEDIUM	MEDIUM: reconstruction from a freed cell whose
urls	FREEBLOCK	id	MEDIUM	MEDIUM: reconstruction from a freed cell whose
visits	FREEBLOCK	id	MEDIUM	MEDIUM: reconstruction from a freed cell whose
visits	FREEBLOCK	id	MEDIUM	MEDIUM: reconstruction from a freed cell whose
visits	FREEBLOCK	id	MEDIUM	MEDIUM: reconstruction from a freed cell whose
visits	FREEBLOCK	id	MEDIUM	MEDIUM: reconstruction from a freed cell whose
visits	FREEBLOCK	id	MEDIUM	MEDIUM: reconstruction from a freed cell whose
visits	FREEBLOCK	id	MEDIUM	MEDIUM: reconstruction from a freed cell whose
visits	FREEBLOCK	id	MEDIUM	MEDIUM: reconstruction from a freed cell whose
visits	FREEBLOCK	id	MEDIUM	MEDIUM: reconstruction from a freed cell whose
visits	FREEBLOCK	id	MEDIUM	MEDIUM: reconstruction from a freed cell whose
visits	FREEBLOCK	id	MEDIUM	MEDIUM: reconstruction from a freed cell whose
visits	FREEBLOCK	id	MEDIUM	MEDIUM: reconstruction from a freed cell whose
visits	FREEBLOCK	id	MEDIUM	MEDIUM: reconstruction from a freed cell whose
visits	FREEBLOCK	id	MEDIUM	MEDIUM: reconstruction from a freed cell whose
visits	FREEBLOCK	id	MEDIUM	MEDIUM: reconstruction from a freed cell whose
visits	FREEBLOCK	id	MEDIUM	MEDIUM: reconstruction from a freed cell whose
sms	UNALLOCATED_	_id	HIGH	HIGH: reconstruction from a freed cell whose hea
sms	FREEBLOCK	_id	HIGH	HIGH: reconstruction from a freed cell whose hea
sms	FREEBLOCK	_id	HIGH	HIGH: reconstruction from a freed cell whose hea
sms	FREEBLOCK	_id	MEDIUM	MEDIUM: reconstruction from a freed cell whose
sms	FREEBLOCK	_id	HIGH	HIGH: reconstruction from a freed cell whose hea
sms	FREEBLOCK	_id	HIGH	HIGH: reconstruction from a freed cell whose hea
sms	FREEBLOCK	_id	HIGH	HIGH: reconstruction from a freed cell whose hea

10. Carved candidates

Every row here is an unconfirmed candidate: status literally CARVED CANDIDATE, never presented as evidence.

Kind	Matched	Container	Confidence	Explanation
sqlite_record	["table", "sqlite_sequence", "sqlite_sequen	sqlite_free_space	LOW	LOW: carved sqlite_record pattern match; no artif
sqlite_record	[null, "+12025550124", 1, 1712107930558	sqlite_free_space	MEDIUM	MEDIUM: carved sqlite_record pattern match; corr
url	content://com.android.contacts/contacts/lo	sqlite_free_space	LOW	LOW: carved url pattern match; no artifact carries
sqlite_record	[null, "+12025550126", 1, 1712714862988	sqlite_free_space	MEDIUM	MEDIUM: carved sqlite_record pattern match; corr
url	content://com.android.contacts/contacts/lo	sqlite_free_space	LOW	LOW: carved url pattern match; no artifact carries
sqlite_record	[null, "+12025550129", 1, 1714905777505	sqlite_free_space	MEDIUM	MEDIUM: carved sqlite_record pattern match; corr
phone_number	+12025550129	sqlite_free_space	MEDIUM	MEDIUM: carved phone_number pattern match; c
sqlite_record	[null, "+12025550120", 1, 1715424030220	sqlite_free_space	MEDIUM	MEDIUM: carved sqlite_record pattern match; corr
url	content://com.android.contacts/contacts/lo	sqlite_free_space	LOW	LOW: carved url pattern match; no artifact carries
sqlite_record	[null, "+12025550127", 1, 1712925289285	sqlite_free_space	MEDIUM	MEDIUM: carved sqlite_record pattern match; corr
url	content://com.android.contacts/contacts/lo	sqlite_free_space	LOW	LOW: carved url pattern match; no artifact carries
sqlite_record	[null, "+12025550127", 1, 1712779140921	sqlite_free_space	MEDIUM	MEDIUM: carved sqlite_record pattern match; corr
url	content://com.android.contacts/contacts/lo	sqlite_free_space	LOW	LOW: carved url pattern match; no artifact carries
sqlite_record	[null, "+12025550121", 1, 1716987759953	sqlite_free_space	MEDIUM	MEDIUM: carved sqlite_record pattern match; corr
url	content://com.android.contacts/contacts/lo	sqlite_free_space	LOW	LOW: carved url pattern match; no artifact carries
sqlite_record	[null, "+12025550106", 1, 1712245251567	sqlite_free_space	MEDIUM	MEDIUM: carved sqlite_record pattern match; corr
url	content://com.android.contacts/contacts/lo	sqlite_free_space	LOW	LOW: carved url pattern match; no artifact carries
sqlite_record	[null, "+12025550126", 1, 1712543387558	sqlite_free_space	MEDIUM	MEDIUM: carved sqlite_record pattern match; corr
url	content://com.android.contacts/contacts/lo	sqlite_free_space	LOW	LOW: carved url pattern match; no artifact carries
sqlite_record	[null, "", 2, 1710548244446, 0, 5, 0, null, n	sqlite_free_space	LOW	LOW: carved sqlite_record pattern match; no artif
sqlite_record	[null, "+12025550124", 1, 1710798999685	sqlite_free_space	MEDIUM	MEDIUM: carved sqlite_record pattern match; corr
phone_number	+12025550124	sqlite_free_space	MEDIUM	MEDIUM: carved phone_number pattern match; c
sqlite_record	[null, "+12025550108", 1, 1709388963631	sqlite_free_space	MEDIUM	MEDIUM: carved sqlite_record pattern match; corr
url	content://com.android.contacts/contacts/lo	sqlite_free_space	LOW	LOW: carved url pattern match; no artifact carries
sqlite_record	[null, "+12025550108", 1, 1715415759483	sqlite_free_space	MEDIUM	MEDIUM: carved sqlite_record pattern match; corr
url	content://com.android.contacts/contacts/lo	sqlite_free_space	LOW	LOW: carved url pattern match; no artifact carries
sqlite_record	[null, "+12025550108", 1, 1714571763568	sqlite_free_space	MEDIUM	MEDIUM: carved sqlite_record pattern match; corr
url	content://com.android.contacts/contacts/lo	sqlite_free_space	LOW	LOW: carved url pattern match; no artifact carries
sqlite_record	[null, "+12025550100", 1, 1710296255564	sqlite_free_space	MEDIUM	MEDIUM: carved sqlite_record pattern match; corr
url	content://com.android.contacts/contacts/lo	sqlite_free_space	LOW	LOW: carved url pattern match; no artifact carries
sqlite_record	[null, "+12025550110", 1, 1710880994968	sqlite_free_space	MEDIUM	MEDIUM: carved sqlite_record pattern match; corr
url	content://com.android.contacts/contacts/lo	sqlite_free_space	LOW	LOW: carved url pattern match; no artifact carries
sqlite_record	[null, "+12025550114", 1, 1709757981963	sqlite_free_space	MEDIUM	MEDIUM: carved sqlite_record pattern match; corr
url	content://com.android.contacts/contacts/lo	sqlite_free_space	LOW	LOW: carved url pattern match; no artifact carries
sqlite_record	[null, "+12025550115", 1, 1716775375947	sqlite_free_space	MEDIUM	MEDIUM: carved sqlite_record pattern match; corr
url	content://com.android.contacts/contacts/lo	sqlite_free_space	LOW	LOW: carved url pattern match; no artifact carries
sqlite_record	[null, "+12025550104", 1, 1711398465101	sqlite_free_space	MEDIUM	MEDIUM: carved sqlite_record pattern match; corr
phone_number	+12025550104	sqlite_free_space	MEDIUM	MEDIUM: carved phone_number pattern match; c
sqlite_record	[null, "+12025550111", 1, 1713648953187	sqlite_free_space	MEDIUM	MEDIUM: carved sqlite_record pattern match; corr
url	content://com.android.contacts/contacts/lo	sqlite_free_space	LOW	LOW: carved url pattern match; no artifact carries

Showing the first 40 of 177 rows. The complete data is in report.json and the CSV files.

11. Timeline

Chronological (UTC). File events are filesystem metadata, not evidence of user activity.

Timestamp	State	Type	Event	Source
2024-03-01T08:28:12.947	PARTIAL	browser_visit	Visited unknown URL	0000000_History ? vis
2024-03-01T17:04:38.000	ACTIVE	contact	Contact Arjun Kapoor ? (202) 555-0110	contacts2.db ? raw_c
2024-03-01T21:11:10.062	ACTIVE	browser_visit	Visited Video #38	History ? visits#34
2024-03-01T22:40:54.056	ACTIVE	browser_visit	Last visit to Forum Thread #29	History ? urls#29
2024-03-01T22:40:54.056	PARTIAL	browser_visit	Visited unknown URL	0000000_History ? vis
2024-03-01T23:30:47.049	ACTIVE	call	Call MISSED from +12025550129 (0s)	calllog.db ? calls#56
2024-03-02T07:06:55.000	ACTIVE	contact	Contact Omar Bose ? (202) 555-0118	contacts2.db ? raw_c
2024-03-02T09:44:48.000	ACTIVE	contact	Contact Yash Rao ? (202) 555-0124	contacts2.db ? raw_c
2024-03-02T11:26:36.255	ACTIVE	browser_visit	Visited Weather #27	History ? visits#76
2024-03-02T13:02:48.814	ACTIVE	sms	SMS RECEIVED from +12025550120: Dinner at my place on	mmssms.db ? sms#5
2024-03-02T14:16:03.631	ACTIVE	call	Call INCOMING from +12025550108 (1651s)	calllog.db ? calls#5
2024-03-02T17:06:11.439	ACTIVE	browser_visit	Visited Weather #27	History ? visits#39
2024-03-03T05:46:24.755	ACTIVE	sms	SMS RECEIVED from 56161: Package out for delivery, arrivin	mmssms.db ? sms#6
2024-03-03T15:44:39.440	ACTIVE	sms	SMS RECEIVED from +12025550108: Can you call me when y	mmssms.db ? sms#5
2024-03-03T18:23:18.707	ACTIVE	sms	SMS DRAFT with unknown address: Meeting moved to room 4	mmssms.db ? sms#1
2024-03-03T18:24:48.149	ACTIVE	browser_visit	Visited Student Portal #36	History ? visits#140
2024-03-03T19:54:30.495	ACTIVE	browser_visit	Visited Inbox #33	History ? visits#77
2024-03-04T02:16:31.205	PARTIAL	browser_visit	Visited unknown URL	0000000_History ? vis
2024-03-04T18:34:47.171	ACTIVE	browser_visit	Visited World News	History ? visits#115
2024-03-04T21:47:24.488	ACTIVE	browser_visit	Visited Shared Doc #20	History ? visits#22
2024-03-05T09:29:05.813	ACTIVE	sms	SMS SENT to +12025550114: Reminder: appointment tomorro	mmssms.db ? sms#1
2024-03-05T20:06:21.718	ACTIVE	call	Call OUTGOING to +12025550102 (757s)	calllog.db ? calls#22
2024-03-06T01:18:20.468	ACTIVE	browser_visit	Visited unknown URL	History ? visits#94
2024-03-06T03:04:47.784	ACTIVE	sms	SMS SENT to +12025550120: Bank alert: Rs 6494 debited fro	mmssms.db ? sms#4
2024-03-06T11:55:49.282	ACTIVE	browser_visit	Visited Video	History ? visits#13
2024-03-06T12:09:01.030	ACTIVE	call	Call OUTGOING to +12025550125 (831s)	calllog.db ? calls#28
2024-03-06T19:42:55.797	ACTIVE	call	Call REJECTED from +12025550113 (0s)	calllog.db ? calls#41
2024-03-06T20:46:21.963	ACTIVE	call	Call REJECTED from +12025550114 (0s)	calllog.db ? calls#40
2024-03-07T01:56:19.552	ACTIVE	sms	SMS SENT to +12025550104: Package out for delivery, arrivin	mmssms.db ? sms#2
2024-03-07T06:04:39.678	ACTIVE	browser_visit	Visited Wikipedia #11	History ? visits#85
2024-03-07T15:29:42.318	ACTIVE	browser_visit	Visited unknown URL	History ? visits#1
2024-03-07T16:44:51.001	ACTIVE	browser_visit	Visited World News #32	History ? visits#51
2024-03-07T21:44:20.006	ACTIVE	browser_visit	Visited World News	History ? visits#8
2024-03-08T02:44:42.000	ACTIVE	contact	Contact Ravi Pillai ? (202) 555-0116	contacts2.db ? raw_c
2024-03-08T04:15:06.000	ACTIVE	contact	Contact Karan Menon ? (202) 555-0112	contacts2.db ? raw_c
2024-03-08T06:58:55.265	ACTIVE	call	Call MISSED from +12025550112 (0s)	calllog.db ? calls#18
2024-03-08T11:06:12.391	ACTIVE	sms	SMS RECEIVED from +12025550106: Payment of Rs 5733 re	mmssms.db ? sms#1
2024-03-08T13:54:41.000	ACTIVE	contact	Contact Isha Nair ? (202) 555-0103	contacts2.db ? raw_c
2024-03-08T15:47:54.903	ACTIVE	browser_visit	Visited Shared Doc #20	History ? visits#91
2024-03-08T23:37:15.314	ACTIVE	sms	SMS RECEIVED from +12025550104: Package out for deliver	mmssms.db ? sms#9

Showing the first 40 of 382 rows. The complete data is in report.json and the CSV files.

12. Correlation findings

Source	Relation	Target	Explanation
database:database:1	FOUND_IN	application:app:com.android.chrome	data/data/com.android.chrome/app_chro
artifact:artifact:1	FOUND_IN	database:database:1	'Visited unknown URL' is a live record of d
artifact:artifact:2	FOUND_IN	database:database:1	'Visited Maps #34' is a live record of data/
artifact:artifact:3	FOUND_IN	database:database:1	'Visited World News #12' is a live record o
artifact:artifact:4	FOUND_IN	database:database:1	'Visited Video #18' is a live record of data/
artifact:artifact:5	FOUND_IN	database:database:1	'Visited Shopping Cart' is a live record of d
artifact:artifact:6	FOUND_IN	database:database:1	'Visited Video #38' is a live record of data/
artifact:artifact:7	FOUND_IN	database:database:1	'Visited World News' is a live record of dat
artifact:artifact:8	FOUND_IN	database:database:1	'Visited Maps #34' is a live record of data/
artifact:artifact:9	FOUND_IN	database:database:1	'Visited World News' is a live record of dat
artifact:artifact:10	FOUND_IN	database:database:1	'Visited Forum Thread' is a live record of d
artifact:artifact:11	FOUND_IN	database:database:1	'Visited Video' is a live record of data/data
artifact:artifact:12	FOUND_IN	database:database:1	'Visited Video #38' is a live record of data/
artifact:artifact:13	FOUND_IN	database:database:1	'Visited Video #38' is a live record of data/
artifact:artifact:14	FOUND_IN	database:database:1	'Visited Shopping Cart' is a live record of d
artifact:artifact:15	FOUND_IN	database:database:1	'Visited Weather #17' is a live record of da
artifact:artifact:16	FOUND_IN	database:database:1	'Visited unknown URL' is a live record of d
artifact:artifact:17	FOUND_IN	database:database:1	'Visited Forum Thread' is a live record of d
artifact:artifact:18	FOUND_IN	database:database:1	'Visited Maps #14' is a live record of data/
artifact:artifact:19	FOUND_IN	database:database:1	'Visited Inbox #33' is a live record of data/
artifact:artifact:20	FOUND_IN	database:database:1	'Visited Shared Doc #20' is a live record of
artifact:artifact:21	FOUND_IN	database:database:1	'Visited Video #28' is a live record of data/
artifact:artifact:22	FOUND_IN	database:database:1	'Visited Weather #17' is a live record of da
artifact:artifact:23	FOUND_IN	database:database:1	'Visited Maps' is a live record of data/data/
artifact:artifact:24	FOUND_IN	database:database:1	'Visited World News' is a live record of dat
artifact:artifact:25	FOUND_IN	database:database:1	'Visited Video #28' is a live record of data/
artifact:artifact:26	FOUND_IN	database:database:1	'Visited Video #18' is a live record of data/
artifact:artifact:27	FOUND_IN	database:database:1	'Visited Inbox #13' is a live record of data/
artifact:artifact:28	FOUND_IN	database:database:1	'Visited unknown URL' is a live record of d
artifact:artifact:29	FOUND_IN	database:database:1	'Visited Shopping Cart #25' is a live record
artifact:artifact:30	FOUND_IN	database:database:1	'Visited Video #38' is a live record of data/
artifact:artifact:31	FOUND_IN	database:database:1	'Visited Student Portal #36' is a live record
artifact:artifact:32	FOUND_IN	database:database:1	'Visited Student Portal #26' is a live record
artifact:artifact:33	FOUND_IN	database:database:1	'Visited Video #28' is a live record of data/
artifact:artifact:34	FOUND_IN	database:database:1	'Visited Inbox #33' is a live record of data/
artifact:artifact:35	FOUND_IN	database:database:1	'Visited Weather #27' is a live record of da
artifact:artifact:36	FOUND_IN	database:database:1	'Visited Student Portal #36' is a live record
artifact:artifact:37	FOUND_IN	database:database:1	'Visited World News #32' is a live record o
artifact:artifact:38	FOUND_IN	database:database:1	'Visited Video #38' is a live record of data/
artifact:artifact:39	FOUND_IN	database:database:1	'Visited Wikipedia' is a live record of data/

Showing the first 40 of 9079 rows. The complete data is in report.json and the CSV files.

13. Validation results

PARTIAL: 12, UNVERIFIED: 8, VALID: 56

Status	Check	Target	Detail
PARTIAL	record_structure	table: data/data/com.android.chrome/app	10 of 150 records have structural/datatype anomalies
PARTIAL	record_structure	artifact: data/data/com.android.chrome/ap	dangling_url_reference: visit points to a url id absent from the
PARTIAL	record_structure	artifact: data/data/com.android.chrome/ap	dangling_url_reference: visit points to a url id absent from the
PARTIAL	record_structure	artifact: data/data/com.android.chrome/ap	dangling_url_reference: visit points to a url id absent from the
PARTIAL	record_structure	artifact: data/data/com.android.chrome/ap	dangling_url_reference: visit points to a url id absent from the
PARTIAL	record_structure	artifact: data/data/com.android.chrome/ap	dangling_url_reference: visit points to a url id absent from the
PARTIAL	record_structure	artifact: data/data/com.android.chrome/ap	dangling_url_reference: visit points to a url id absent from the
PARTIAL	record_structure	artifact: data/data/com.android.chrome/ap	dangling_url_reference: visit points to a url id absent from the
PARTIAL	record_structure	artifact: data/data/com.android.chrome/ap	dangling_url_reference: visit points to a url id absent from the
PARTIAL	record_structure	artifact: data/data/com.android.chrome/ap	dangling_url_reference: visit points to a url id absent from the
PARTIAL	sqlite_header	database: data/data/com.example.corrupt/	header valid but file size is inconsistent: header declares 12 p
UNVERIFIED	sqlite_header	database: data/data/com.example.secure/	database-like name without SQLite header; high entropy (7.96
UNVERIFIED	sqlite_integrity	database: data/data/com.example.corrupt/	database could not be opened by SQLite
UNVERIFIED	sqlite_integrity	database: data/data/com.example.secure/	not attempted: file appears encrypted
UNVERIFIED	timestamp_plausibility	table: data/data/com.android.chrome/app	records carry no timestamp to check
UNVERIFIED	timestamp_plausibility	table: data/data/com.android.providers.co	records carry no timestamp to check
UNVERIFIED	timestamp_plausibility	table: data/data/com.android.providers.tel	records carry no timestamp to check
UNVERIFIED	timestamp_plausibility	table: data/data/com.example.corrupt/data	records carry no timestamp to check
UNVERIFIED	timestamp_plausibility	table: data/data/com.example.notes/datab	records carry no timestamp to check
VALID	acquisition_completeness	source: source s1 android_dump	all 16 entries acquired; none skipped
VALID	cross_artifact_consistenc	case: call log vs contacts	cached call names agree with contacts; 59 of 60 calls have a
VALID	cross_artifact_consistenc	case: SMS thread consistency	every SMS thread maps to a single address
VALID	custody_chain	case: chain-of-custody log	custody chain intact
VALID	duplicate_detection	case: sms artifacts	no duplicate sms records among 120
VALID	duplicate_detection	case: call artifacts	no duplicate call records among 60
VALID	duplicate_detection	case: browser_visit artifacts	no duplicate browser_visit records among 156
VALID	duplicate_detection	case: contact artifacts	no duplicate contact records among 30
VALID	filename_schema_agree	database: data/data/com.android.chrome/	file name and content agree (156 browser_visit artifacts)
VALID	filename_schema_agree	database: data/data/com.android.provider	file name and content agree (60 call artifacts)
VALID	filename_schema_agree	database: data/data/com.android.provider	file name and content agree (30 contact artifacts)
VALID	filename_schema_agree	database: data/data/com.android.provider	file name and content agree (120 sms artifacts)
VALID	hash_consistency	source: source s1 android_dump	all 16 original files re-hashed and match ingest-time SHA-256
VALID	hash_consistency	case: preserved evidence copies	all 13 write-protected preserved copies match their recorded S
VALID	manifest_integrity	case: evidence_manifest.json	manifest matches the hash recorded in the custody log
VALID	provenance_completenes	case: artifact provenance	all 8590 artifacts trace to an evidence file, SHA-256, table and
VALID	record_structure	table: data/data/com.android.chrome/app	6 records: field types consistent with the schema
VALID	record_structure	table: data/data/com.android.chrome/app	5 records: field types consistent with the schema
VALID	record_structure	table: data/data/com.android.providers.co	60 records: field types consistent with the schema
VALID	record_structure	table: data/data/com.android.providers.co	30 records: field types consistent with the schema

Showing the first 40 of 76 rows. The complete data is in report.json and the CSV files.

14. Recovery methodology

Evidence integrity

Every input file is read through a read-only handle exactly once and hashed with SHA-256 (integrity anchor), SHA-1 and MD5 (legacy reference only). Only SQLite databases, their sidecars and small text/XML/JSON files are copied into a write-protected (mode 0444) case evidence store. Originals are never opened for writing and are re-hashed during validation. A hash-chained chain-of-custody log commits each entry to the previous one.

Filesystem analysis

Files are classified by content signature first and file name second (SQLite header magic, WAL/journal magic, XML/ABX, media signatures). The Android directory layout (/data/data, /data/user/N, /sdcard, /storage/emulated/N, ?) is detected from the evidence rather than assumed; application packages are inferred from paths. Symlinks and special files are never followed.

SQLite discovery

The 100-byte SQLite header is parsed in pure Python (page size, encoding, page count, freelist count, schema cookie, journal mode) and checked against the file size. WAL, shared-memory and rollback-journal files are associated with their database and their headers recorded. Each database is opened read-only on a disposable scratch copy so that WAL replay or hot-journal rollback never touches the preserved evidence. PRAGMA integrity_check is recorded.

Active artifact extraction

Tables are identified by schema similarity, not file name. Parsers score how many of the columns they understand are present; that score is stored with every artifact. Fields absent from a schema variant are marked UNKNOWN rather than guessed, and are distinct from stored NULLs.

Normalisation

Timestamps are converted from their declared storage format (Unix ms, WebKit µs, ?) to UTC with the raw value and a plausibility verdict retained. Phone numbers keep their original text and receive a conservative matching key (trailing ten digits of a genuine phone number; short codes and alphanumeric senders receive none).

Validation

Automated checks cover hash consistency, custody chain, SQLite integrity, schema agreement, record structure and datatypes, timestamp plausibility, duplicate detection, cross-artifact consistency and provenance completeness. Failed checks are kept and explained.

Recovery, carving and correlation (this review)

Every page image (the main file's pages by role, WAL frames, rollback-journal pages) is scanned for byte sequences that parse as a record of a known table, excluding live cell extents; a freed cell whose first bytes were overwritten by SQLite's own freeblock header is reconstructed from the surviving bytes with the lost column(s) marked UNKNOWN, never guessed. Raw carving finds pattern candidates (phone numbers, URLs, e-mail addresses, timestamps, SMS-like text, loose SQLite-record shapes) in bytes no parser reads; every hit is a CARVED CANDIDATE, never confirmed evidence, until validated and corroborated. Correlation links artifacts and candidates that share a fact (a phone number, a URL, a timestamp, message text) across independent sources ? a different database, never the same file's own data seen twice. Confidence is a fixed, explainable, deterministic calculation (schema match, structural validity, provenance, corroboration, completeness, timestamp plausibility); the same bytes always score the same way, and a carved candidate can never be HIGH.

15. Limitations

Recoverability of deleted or superseded data depends on the acquisition method and completeness, the filesystem and storage medium (including flash TRIM / garbage collection), the state of each SQLite database (auto-vacuum, VACUUM, checkpointing, journal mode, page reuse), device or file-based encryption, subsequent overwriting, the Android version, and the behaviour of each application. The absence of recovered data in this report is not evidence that data never existed, and the presence of data is only as reliable as the provenance shown beside it. This tool supports, and does not replace, the judgement of a qualified examiner. Use only on evidence you are legally authorised to examine.

- Deleted-data recovery, raw carving and cross-artifact correlation WERE performed for this report (sections 7, 9, 10 and 12). Recoverability still depends on everything the disclaimer above describes ? acquisition completeness, filesystem behaviour, database state, encryption, overwriting: the absence of a recovered record here is not evidence that it never existed.

- 'Active' means present in the database as SQLite itself presents it: main file plus committed WAL frames, with any hot rollback journal rolled back on a scratch copy. Rows flagged deleted by an application (for example Android contacts with deleted=1) are shown as active with a marked-deleted flag.

- Databases that SQLite refuses to open (truncated, malformed, encrypted) yield no artifacts in Review 1. Salvage of damaged databases belongs to the Review 2 page-level engine.

- Encrypted databases are detected and reported but never decrypted; no credential recovery or lock bypass is attempted.

- MMS attachments, messaging apps other than the AOSP SMS provider, and app-specific schemas are catalogued generically (schema, row counts, column hints) but not interpreted.

- Android Binary XML (ABX, Android 12+) files are detected but not decoded. SharedPreferences XML contributes key names only; values are deliberately not extracted.

- Zip archives store local time without a timezone; those times are labelled as such and shown as UTC. File-system times reflect the acquisition method (extraction may reset them).

- Phone-number matching uses a heuristic key, not full E.164 parsing with a numbering-plan database.

- Correctness of the parsers was verified against deterministic synthetic evidence with known ground truth; real devices vary by vendor and Android version.

16. Appendix

A. Skipped / flagged entries

Stage	Issue	Path	Detail
scan	database_truncated	data/data/com.example.corrupt/databases/bro	29437 bytes on disk but the header implies 49152
scan	encrypted_suspect	data/data/com.example.secure/databases/vau	database-like name without SQLite header; high entro
scan	binary_xml_not_parsed	data/system/packages.xml	Android Binary XML (ABX); not parsed in Review 1
scan	database_unopenable	data/data/com.example.corrupt/databases/bro	SQLite refused to open it: DatabaseError: database di

B. Chain of custody

VALID: custody chain intact

C. Analysis runs

#	Stage	Status	Started	Finished
1	scan	COMPLETED	2026-09-22T07:48:31.398Z	2026-09-22T07:48:31.424Z
2	scan	COMPLETED	2026-09-22T07:48:39.443Z	2026-09-22T07:48:39.464Z
3	extract	COMPLETED	2026-09-22T07:48:39.466Z	2026-09-22T07:48:39.511Z
4	recover	COMPLETED	2026-09-22T07:48:39.512Z	2026-09-22T07:48:45.535Z
5	carve	COMPLETED	2026-09-22T07:48:45.548Z	2026-09-22T07:48:46.081Z
6	correlate	COMPLETED	2026-09-22T07:48:46.094Z	2026-09-22T07:48:47.712Z
7	reconstruct	COMPLETED	2026-09-22T07:48:47.720Z	2026-09-22T07:48:54.430Z
8	validate	COMPLETED	2026-09-22T07:48:54.432Z	2026-09-22T07:48:54.572Z
9	timeline	COMPLETED	2026-09-22T07:48:54.573Z	2026-09-22T07:48:54.584Z
10	validate	COMPLETED	2026-09-22T07:49:15.511Z	2026-09-22T07:49:15.688Z

D. Status glossary

ACTIVE: Record present in the live database view. Read through the SQLite engine on a scratch copy.

RECOVERED / PARTIAL / CARVED / CORRUPTED / UNRECOVERABLE / INVALID: Artifact statuses reserved for the Review 2 recovery engine; none are produced in Review 1. A CARVED artifact is a candidate, never confirmed evidence.

VALID: The check passed.

PARTIAL: The check passed with reservations that are explained in the result.

INVALID: The check failed. Failures are kept and explained, never discarded.

UNVERIFIED: The check could not be performed (for example the original evidence is no longer available, or the database is encrypted).